Homework 29:

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CS-F

Problem 1:

$F_n - F_{n-1} - F_{n-2} = 0$

$F_n = d^n$

$d^n = d^{n-1} + d^{n-2}$

Divide by d^{n-2}

$d^2 = d + 1$

$d^2 - d - 1 = 0$

$d_1 = \frac{1+\sqrt{5}}{2}$ or $d_2 = \frac{1-\sqrt{5}}{2}$

$\phi = \frac{1+\sqrt{5}}{2}$ or $\bar{\phi} = \frac{1-\sqrt{5}}{2}$

$F_n = c_1 d_1^n + c_2 d_2^n$

$F_0 = 0$ & $F_1 = 1$

For $F_0 = 0$

$F_0 = c_1 d_1^0 + c_2 d_2^0$

$0 = c_1 + c_2$

$c_1 = -c_2$

For $F_1 = 1$

$F_1 = c_1 d_1 + c_2 d_2$

$= c_1 d_1 + (-c_1)(d_2)$

$= c_1 (d_1 - d_2)$

Since $d_1 - d_2 = \sqrt{5}$

$C_1\sqrt{5} = 1$

$C_1 = \frac{1}{\sqrt{5}}$ and $C_2 = -\frac{1}{\sqrt{5}}$

$$F_n = \frac{1}{\sqrt{5}}\left(\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right)$$

Problem 2:

$x_0 = 1, \quad x_1 = 6$

$x_n = d^n$

$d^n = 6d^{n-1} - 9d^{n-2}$

$d^2 = 6d - 9$

$d^2 - 6d + 9 = 0$

$d_2 = d_1 = 3$ ~~and d~~

$x_n = c_1 d^n + c_2 n d^n$

$x_n = c_1 3^n + c_2 n 3^n$

For $x_0 = 1$

$1 = c_1 3^0 + c_2 (0) 3^1$

$1 = c_1$

For $x_1 = 6$

$6 = 1 \cdot 3^1 + c_2 (1) 3^1$

$6 = 3 + 3c_2$

$c_2 = 1$

Thus, $x_n = 3^n + n3^n$

$x_n = (n+1)3^n$